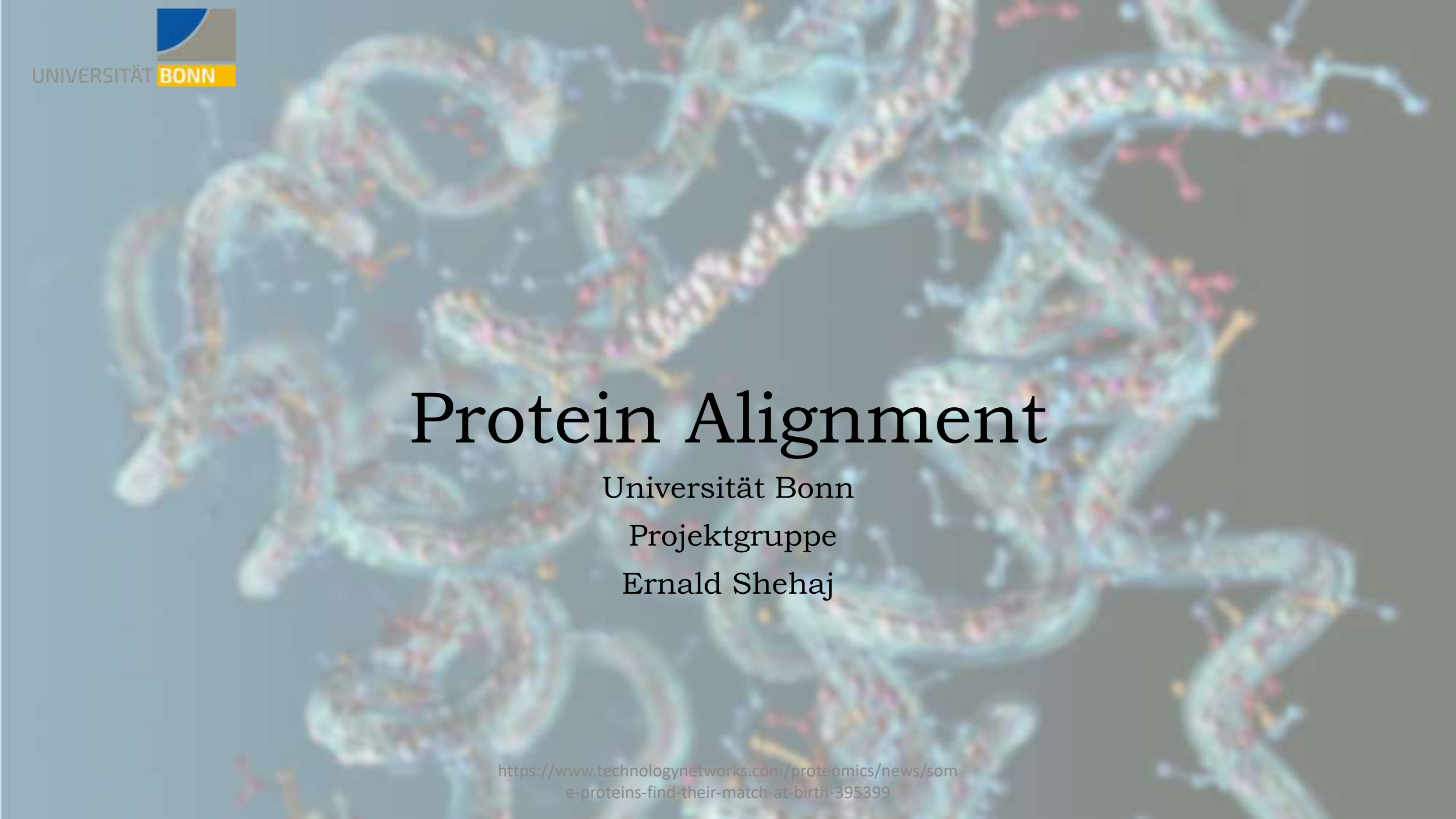


Protein Alignment

Universität Bonn

Projektgruppe

Ernald Shehaj

https://www.technologynetworks.com/proteomics/news/some-proteins-find-their-match-at-birth-395399

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What is Protein alignment?

- The process of comparing two or more DNA, RNA, or protein sequences to identify similarities between them.
- These similarities give hints into:
  - the functional regions of a protein
  - important structural characteristics of sequences
  - the evolutionary relationship between sequences.

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Different categories of methods based on protein alignment

- shape retrieval methods based on histograms summarizing local or global geometrical features of the surface shape.
- shape retrieval methods based on molecular surface maps, i.e. the projection(s) of the protein topography in the 3D space.
- Feature-based 3D protein alignment

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Different categories of methods based on protein alignment

- shape retrieval methods based on histograms summarizing local or global geometrical features of the surface shape.

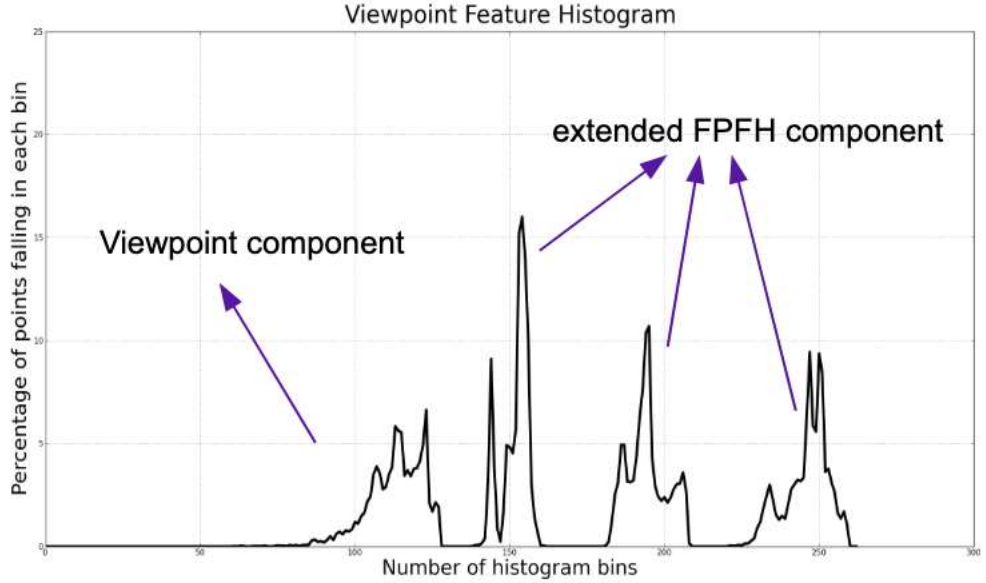


Fig. 6. An example of the resultant Viewpoint Feature Histogram for one of the objects used. Note the two concatenated components.



Fig. 7. The turn table used to collect views of objects with known orientation.

http://vigir.missouri.edu/~gdesouza/Research/Conference_CDs/IEEE_IROS_2010/data/papers/0497.pdf

Different categories of methods based on protein alignment

- shape retrieval methods based on molecular surface maps, i.e., the projection(s) of the protein topography in the 2D space.

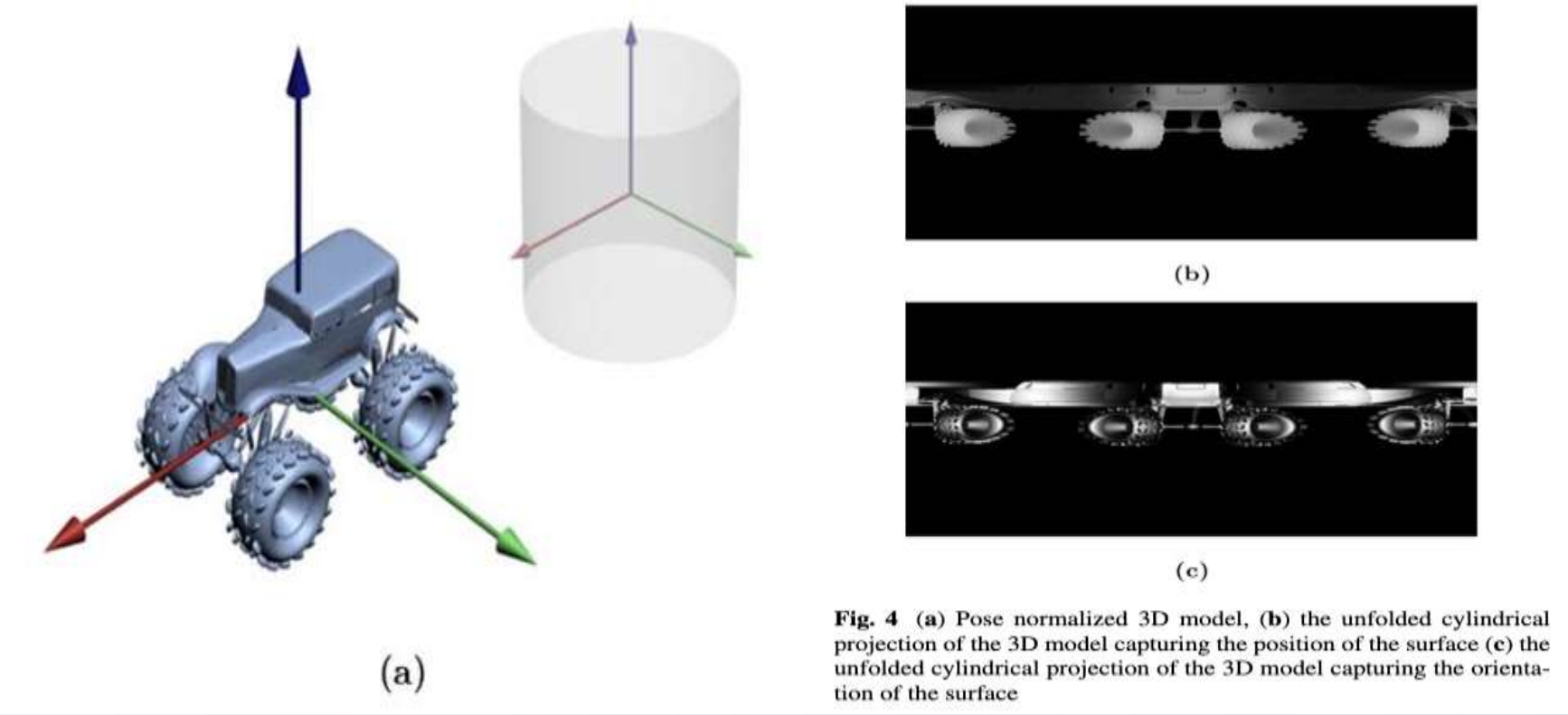


Fig. 4 (a) Pose normalized 3D model, (b) the unfolded cylindrical projection of the 3D model capturing the position of the surface (c) the unfolded cylindrical projection of the 3D model capturing the orientation of the surface

https://scispace.com/pdf/panorama-a-3d-shape-descriptor-based-on-panoramic-views-for-38r1izn6bz.pdf

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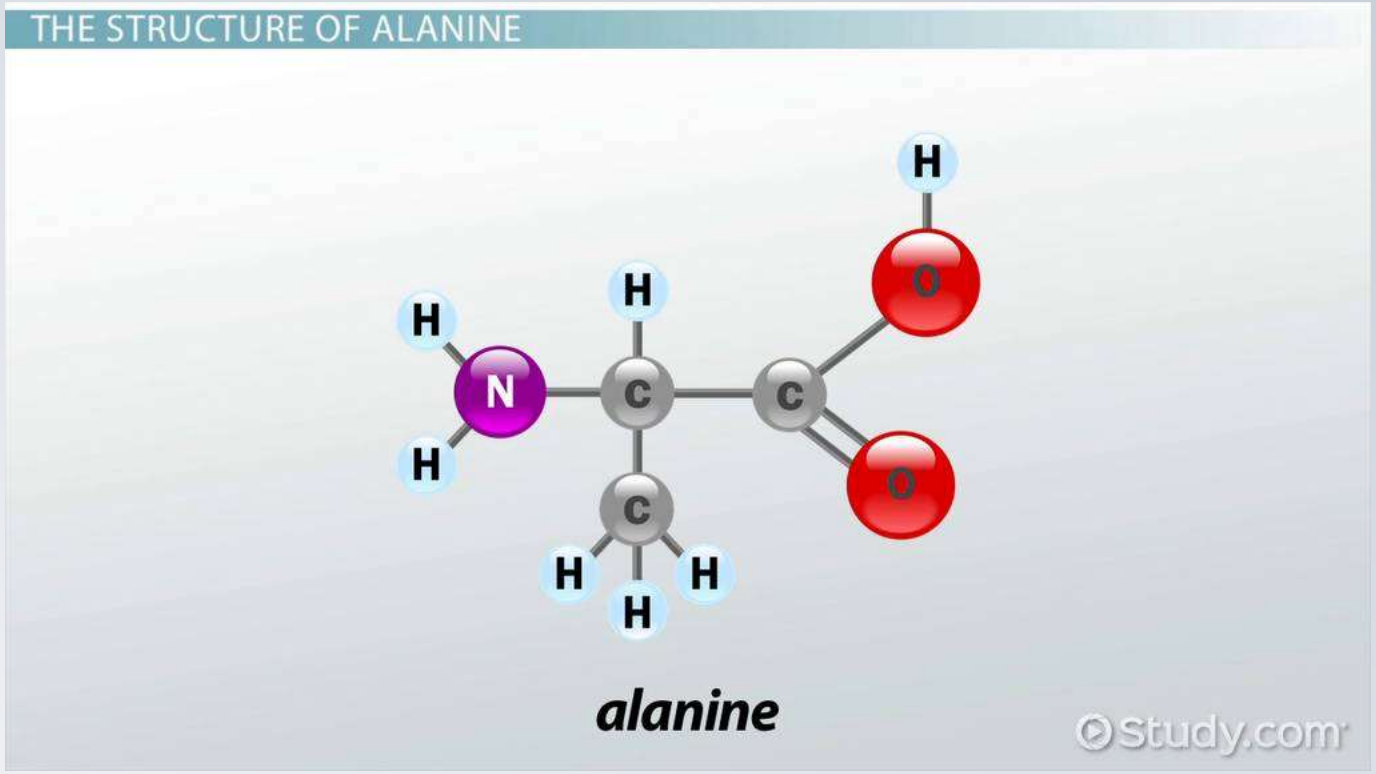
What are Proteins?

- Consisting of a linear sequence of amino acids(what is an amino acid) each linked to its neighbor.
- **Disulfide bonds** between cysteine residues provide additional structural stability beyond peptide bonds.
- The Shape of a Protein Is Specified by Its Amino Acid Sequence
- Large proteins are composed of several discrete regions, known as **chains**

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What are Proteins?

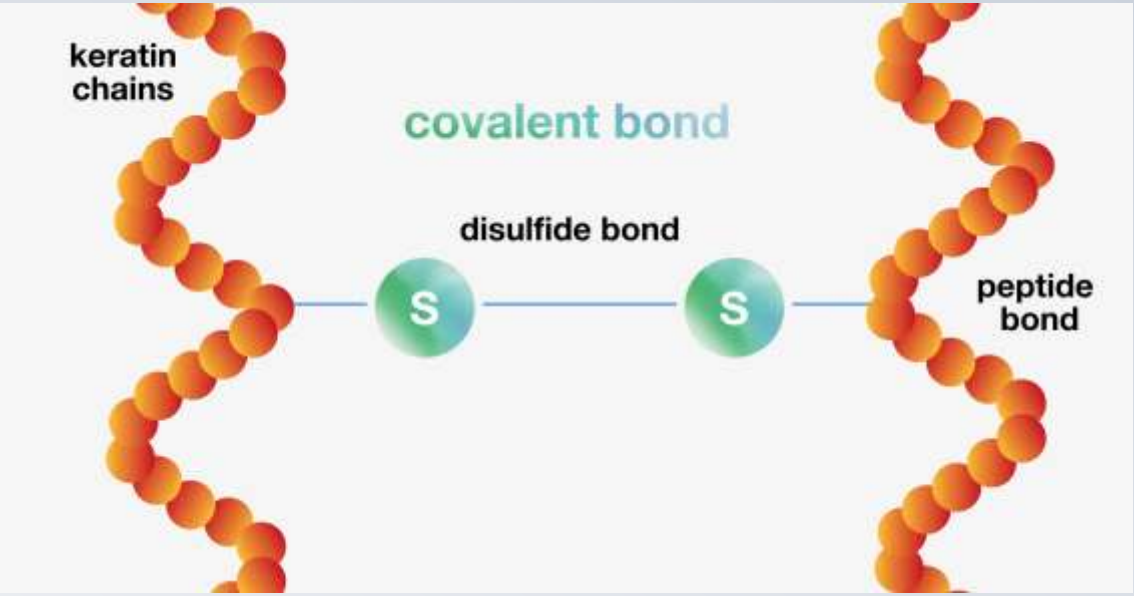
- Consisting of a linear sequence of amino acids each linked to its neighbor.



https://study.com/academy/lesson/video/alanine-structure-formula-quiz.html

What are Proteins?

- **Disulfide bonds**

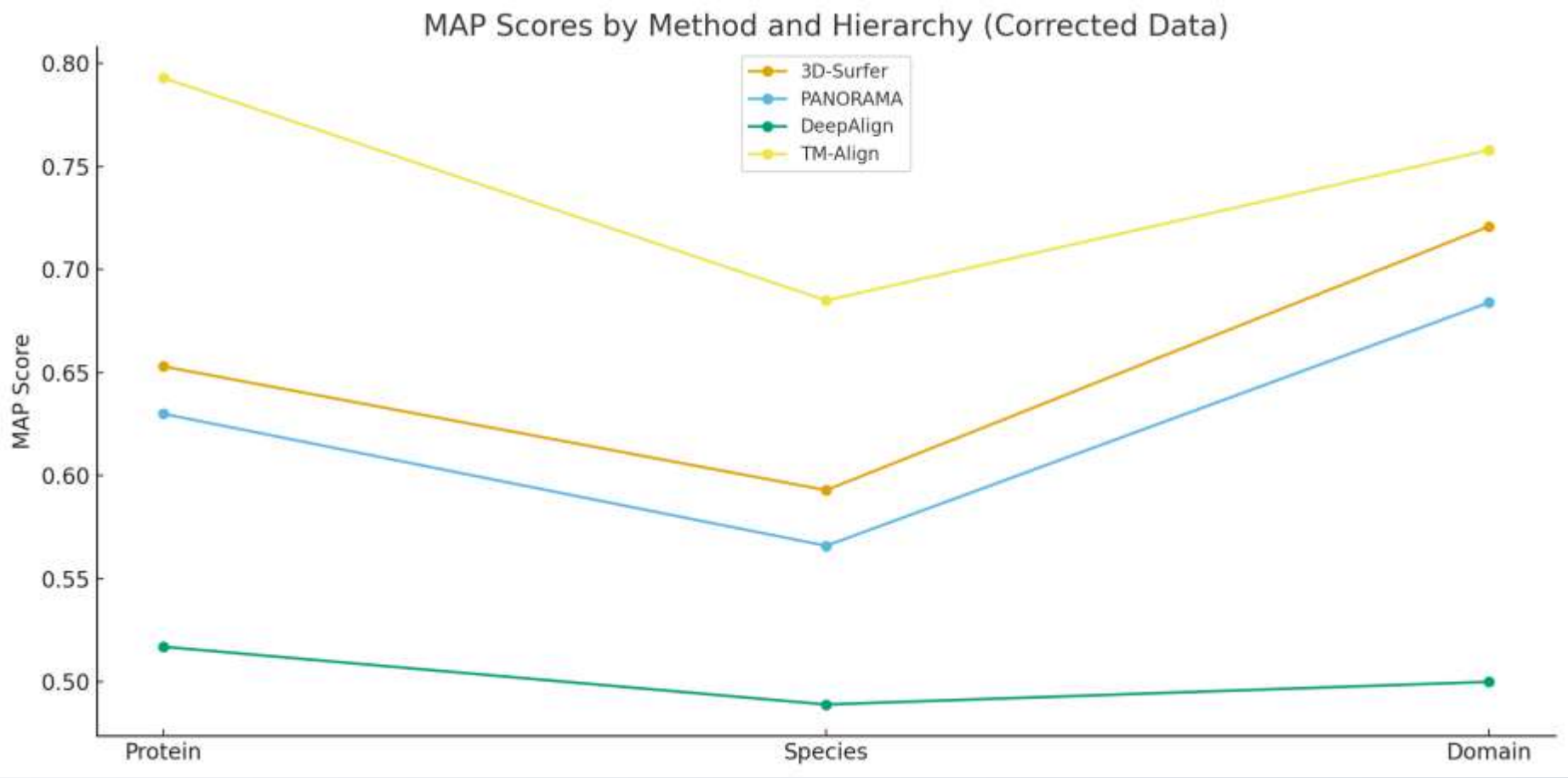


https://www.k18hair.com/blogs/consumer/science-class-13-bonds-holistic-hair-health?srsltid=AfmBOoqcAa9yftZahwcLncn3vKkzR07Jw3EQZJ0bgucGrV7wx3Y5pwjl

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Related work

Comparative Evaluation of Shape Retrieval Methods on Macromolecular Surfaces



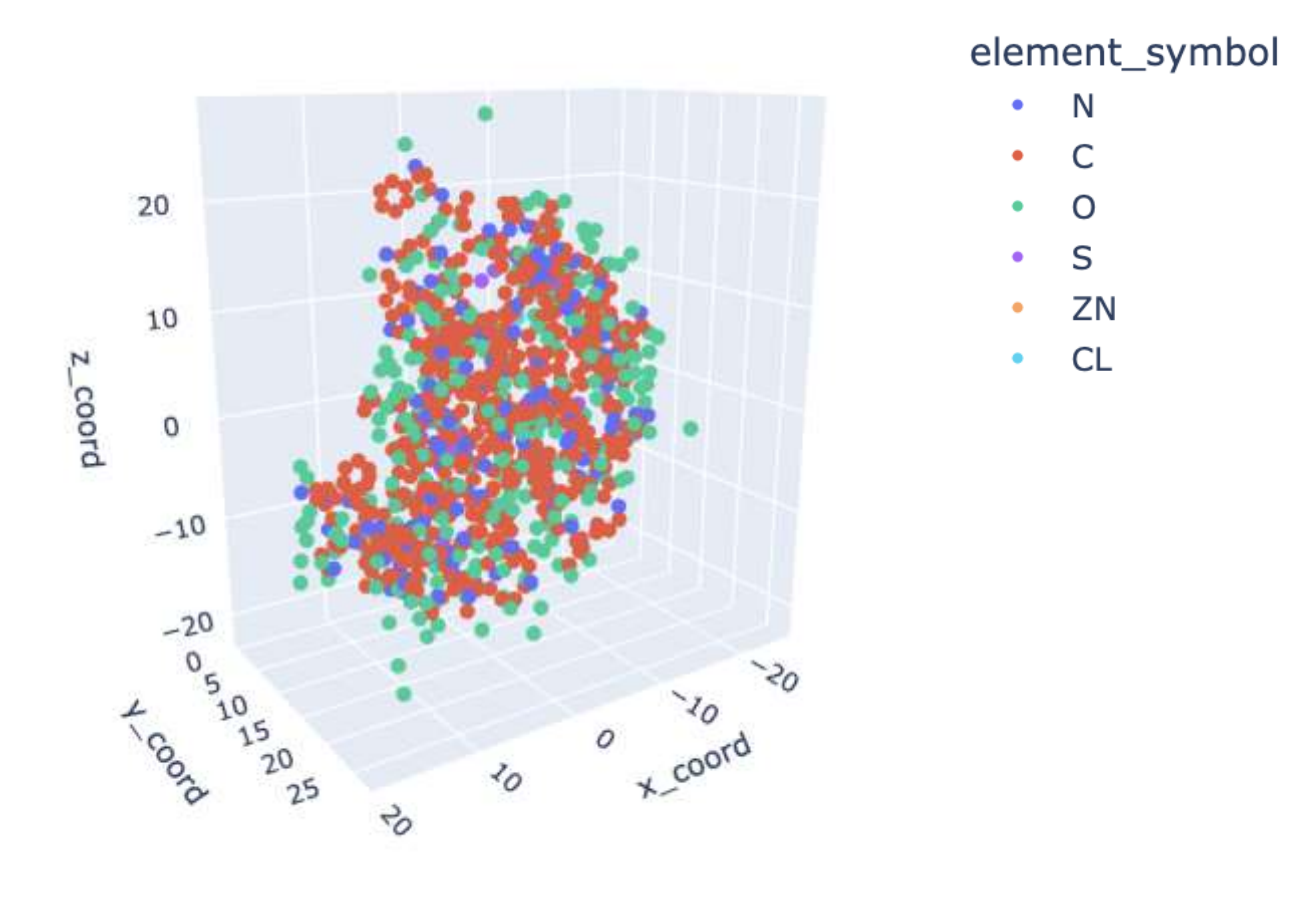
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PDB Files

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HETATM  924  O   HOH D  54      -0.728   2.757  -2.902  1.00 52.70           O
HETATM  925  O   HOH D  55       7.436  20.464  15.487  1.00 64.27           O
HETATM  926  O   HOH D  56       4.515   4.955  -4.053  1.00 48.23           O
HETATM  927  O   HOH D  57      16.146   7.236 -14.395  1.00 66.22           O
HETATM  928  O   HOH D  58       0.004  -0.004 -11.812  0.33 82.86           O
HETATM  929  O   HOH D  59       2.215   3.633  -2.334  1.00 42.98           O
HETATM  930  O   HOH D  60      -5.844  14.916 -13.781  1.00 59.32           O
HETATM  931  O   HOH D  61      14.641   4.689 -10.069  1.00 40.77           O
HETATM  932  O   HOH D  62      11.550  15.703   7.081  1.00 53.44           O
HETATM  933  O   HOH D  63     -10.965  12.514 -12.677  1.00 59.24           O
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CONECT   49  223
CONECT   76   43
CONECT  154  318
CONECT  223   49
CONECT  247  826
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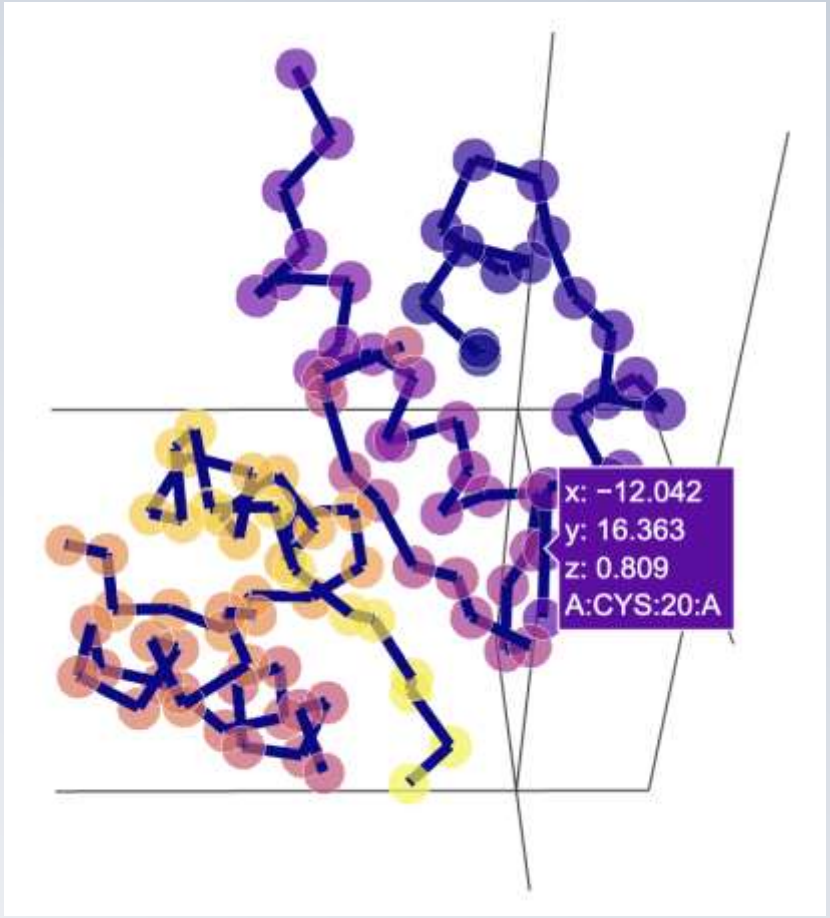
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Plotted Atoms(Pig insuline)



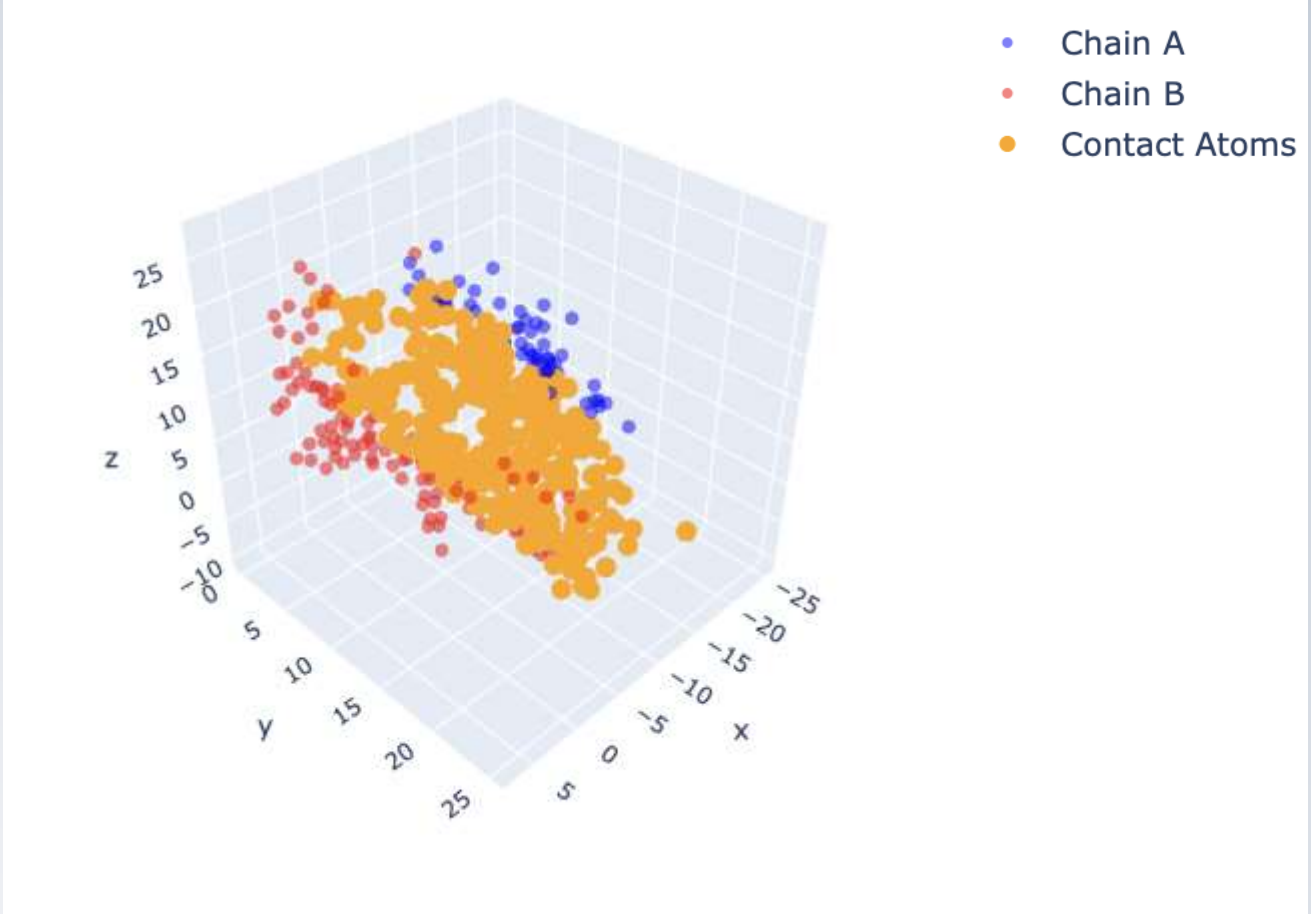
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Backbone Protein Graph



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Chain Interaction

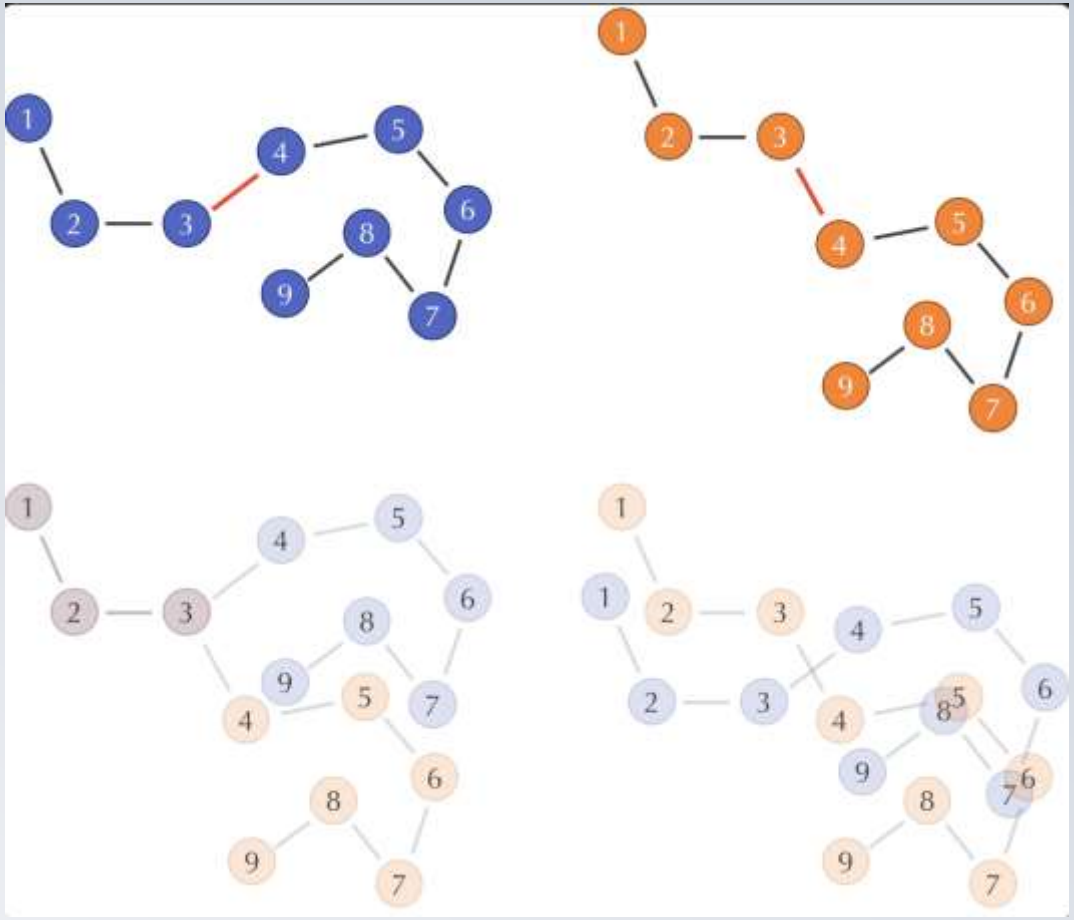


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Feature-based 3D protein alignment

- Cα distances (kNN fingerprints)
- Backbone angles&dihedrals
- Secondary structure (helix/strand/coil)
- Residue chemistry (hydrophobic, polar, charged)
- Spatial fit (local RMSD)

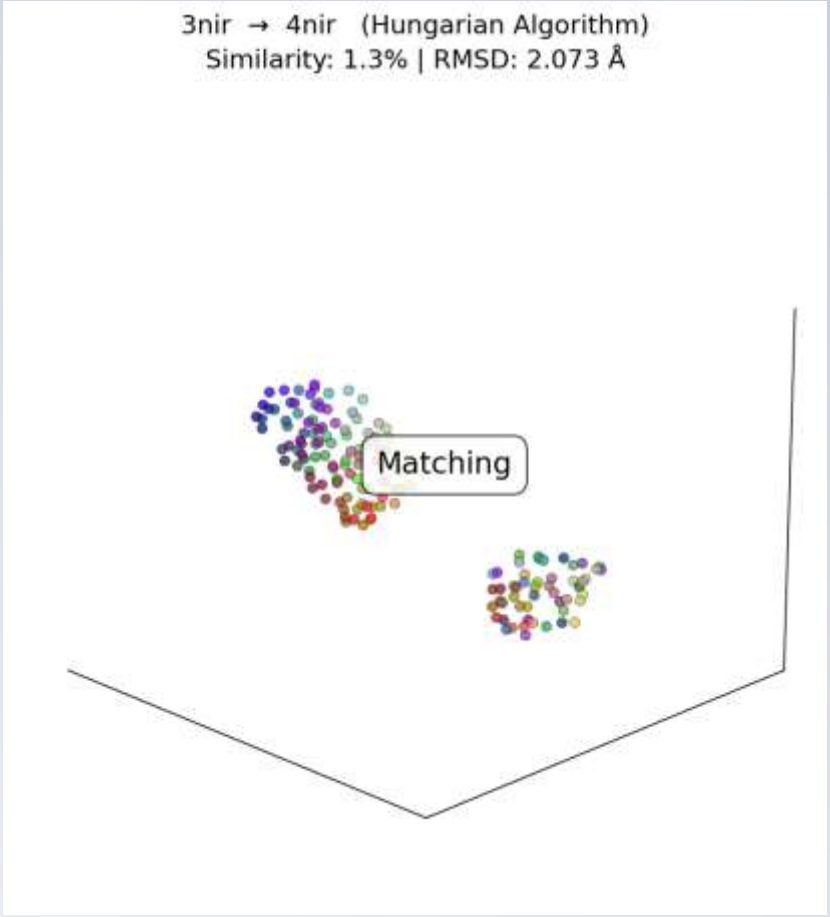
Kabsch-Algorithm



https://hunterheidenreich.com/posts/kabsch-algorithm/

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Feature-based 3D protein alignment



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Feature-based 3D protein alignment vs TM-Align

```
 ************************************************************************
 *                        TM-align (Version 20190822)                   *
 * An algorithm for protein structure alignment and comparison          *
 * Based on statistics:                                                 *
 *       0.0 < TM-score < 0.30, random structural similarity            *
 *       0.5 < TM-score < 1.00, in about the same fold                  *
 * Reference: Y Zhang and J Skolnick, Nucl Acids Res 33, 2302-9 (2005)  *
 * Please email your comments and suggestions to: zhanglab@zhanggroup.org *
 ************************************************************************

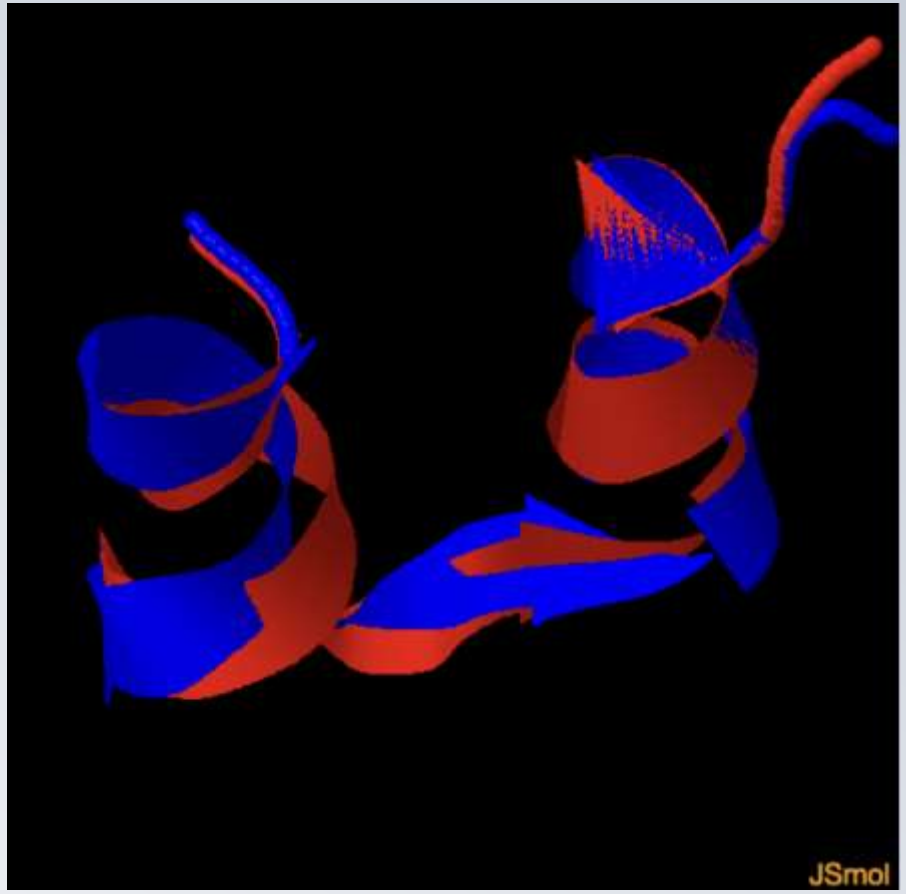
Name of Chain_1: A999395
Name of Chain_2: B999395
Length of Chain_1:   21 residues
Length of Chain_2:   21 residues

Aligned length=   21, RMSD=   1.11, Seq_ID=n_identical/n_aligned= 1.000
TM-score= 0.40069 (if normalized by length of Chain_1)
TM-score= 0.40069 (if normalized by length of Chain_2)
```

Y. Zhang, J. Skolnick, TM-align: A protein structure alignment algorithm based on TM-score,Nucleic Acids Research, 2005 33: 2302-2309.

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Feature-based 3D protein alignment vs TM-Align



Y. Zhang, J. Skolnick, TM-align: A protein structure alignment algorithm based on TM-score,Nucleic Acids Research, 2005 33: 2302-2309.

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Conclusion

- Feature-based comparison
- Rotation-invariant
- Flexible size handling
- Similarity scoring
- Captures overall 3D fold
- Visual output

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What is left to do?

- Sequence info
- Gap handling
- Greedy Matching
- Reducing noise sensitivity

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- “Secondary structure assignment was approximated from Cα geometry using virtual bond angles (80°–120° for helices, 100°–140° for sheets) and virtual dihedrals (–90° to –30° for helices, ±160° for sheets), following typical ranges reported by Levitt (1976) and summarized by Kolodny et al. (2002) and Finkelstein&Ptitsyn (2016).”